Austin J. Baird, PhD

Research Assistant Professor · Department of Surgery | Division of Healthcare Simulation Sciences
826 NE 84th Street, Seattle Wa, 98115
(801) 792-2582 | ajbaird86@gmail.com | bairdlab.org | ajbaird | bairdaustin | Austin Baird

About

My research looks at developing computational models of human biology that span multiple spatial scales, from cellular function to whole-body physiological responses. My goal is to positively impact quality of healthcare and patient safety through collaborative research. Team oriented, empathetic and goal driven.

Education

University of California, Santa Cruz — *Santa Cruz, CA*
BA in Mathematics — *September 2004 - June 2008*
- *Honors Thesis:* Modeling Native California Grassland Populations

University of North Carolina, Chapel Hill — *Chapel Hill, NC*
PhD in Applied Mathematics — *August 2009 - August 2014*
- *Thesis:* Modeling Valveless Pumping Mechanisms, link
- *Advisor:* Dr. Laura Miller, *Committee:* Forest, M. Gregory, Adalsteinsson, David, White, Brian, Mucha, Peter, Kier, William M.

Faculty Positions Held

Duke University — *Durham, NC*
Visiting Assistant Professor — *August 2014 - July 2015*

University of Washington — *Seattle, Wa*
Research Assistant Professor — *December 2021 - Present*

Honors

University of North Carolina, Chapel Hill — *Chapel Hill, NC*
Future Faculty Fellow — *June 2012*

University of North Carolina, Chapel Hill — *Chapel Hill, NC*
J. Burton Linker Award in Recognition of Excellence in Teaching — *March 2013*

University of North Carolina, Chapel Hill — *Chapel Hill, NC*
Graduate Assistance in Areas of National Need Fellow — *Fall 2013*

Applied Research Associates, Inc. — *Raleigh, NC*
Above and Beyond the Call of Duty Award — *October 2016*

Department of Defense — *Washington D.C.*
Federal IT Innovations Award — *August 2017*

Applied Research Associates, Inc. *Raleigh, NC*
DISTINGUISHED MEMBER OF THE TECHNICAL STAFF *June 2018*

Applied Research Associates, Inc. *Raleigh, NC*
TECHNICAL EXCELLENCE AWARD *September 2019*

Affiliations

Society for Simulation in Healthcare
LEADERSHIP GROUP ON SIMULATION AND MODELING *2016 - Present*

IEEE
ENGINEERING IN MEDICINE AND BIOLOGY *2018 - Present*

NIH
INTER-AGENCY MODELING AND ANALYSIS GROUP *2019 - Present*

Teaching Experience

Champion School *San Jose, CA*
COMPUTER SCIENCE, MATH, AND PHYSICS MIDDLE SCHOOL INSTRUCTOR *August 2008*
- Full course responsibility

University of North Carolina, Chapel Hill, Math 190 *Chapel Hill, NC*
CHAOS AND POPULATION DYNAMICS *Fall 2012*
- Full course responsibility

University of North Carolina, Chapel Hill, Math 290 *Chapel Hill, NC*
APPLIED MATHEMATICAL METHODS (COMPUTATIONAL LAB) *Spring 2013*
- Full course responsibility
- *Course Materials:* link

Duke University, Math 353 *Durham, NC*
INTRODUCTION TO ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS FOR ENGINEERS *Fall 2014*
- Full course responsibility

Duke University, Math 353 *Durham, NC*
INTRODUCTION TO ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS FOR ENGINEERS *Spring 2015*
- Full course responsibility

Mentees *Raleigh, NC*
APPLIED RESEARCH ASSOCIATES *Spring 2016 - October 2021*
- Laura Oelsner
  Dates: May 2018 - August 2019
  Current Position: Becton Dickerson Biomedical Engineer
- Tim Hoer
  Dates: May 2018 - August 2018
  Current Position: Fuel Cell System Engineer
- Rishi Das
  Dates: May 2020 - August 2020
  Current Position: Harvard Medical School Engineering Internship

Seattle, WA

University of Washington — *Oct 2022 Present*

- Namrata Harish
  Dates: Oct 2022 - Present
  Current Position: BioEngineering Student, University of Washington

Editorial Responsibilities

Frontiers in Physiology: Computational Physiology and Medicine

Review Editor — *June 2022 - Present*

National Responsibilities

Society of Simulation in Healthcare

Chair: Healthcare Systems Modeling Affinity Group — *March 2023 - Present*

Local Responsibilities

University of Washington — *Seattle, Wa*

Faculty Senate-Counsil on Cyber Security — *September 2023 - Present*

Funding

Defense Health Agency — *Fredrick, MD*

BioGears Research Award — *August 2013 - August 2018*

- *Role:* Principal Investigator
- *Amount:* $7,800,000
- *Contract(s):* W81XWH-13-2-0068,

Defense Health Agency — *Fredrick, MD*

BioGears Follow-on Research Award — *August 2017 - August 2020*

- *Role:* Principal Investigator
- *Amount:* $1,900,000
- *Contract(s):* W81XWH-17-C-0172

Army Research Labs — *Raleigh, NC*

Fast Computational Simulations of Traumatic Brain Injury — *January 2017 - December 2020*

- *Role:* Principal Investigator
- *Amount:* $353,000
- *Contract(s):* W911NF-17-1-0572

Defense Health Agency — *Orlando, FL*

BurnCare: Virtual Patient Application to Train Thermal Injury — *January 2018 - Dec 2021*

- *Role:* Principal Investigator
- *Amount:* $2,100,000
- *Contract(s):* W911NF-18-C-0037

Defense Health Agency — *Fredrick, MD*

SUSTAIN: Prolonged Field Care Training Framework — *March 2018 - Dec 2021*

- *Role:* Lead Physiology Modeler and Proposal Manager
- *Amount:* $2,200,000
- *Contract(s):* W81XWH-18-C-0169

CyberPatient *Vancouver, BC*
CyberPatient: Technical Consultations *Oct 2022 - Jan 2023*
- *Role:* PI
- *Amount:* $23,910

Regis University *Denver, CO*
Overwatch AI *March 2023 - Present*
- *Role:* Co-PI
- *Amount:* $68,782.59

VCom3D *Orlando, Fl*
EXTREME-Physiological Models and Training *September 2023 - Present*
- *Role:* PI
- *Amount:* $160,000.00

Publications

First Section: Peer Reviewed Articles

[1] **Modeling Valveless Pumping Mechanisms**
A. Baird
College of Arts and Sciences, Department of Mathematics (2014)

[2] **Neuromechanical Pumping: Boundary Flexibility and Traveling Depolarization Waves Drive Flow Within Valveless, Tubular Hearts**
A. Baird, L. Waldrop, L. Miller
Japan Journal of Industrial and Applied Mathematics 32.3 (2015) pp. 829–846 Springer

[3] **A Mathematical Model and MATLAB Code for Muscle–Fluid–Structure Simulations**
N. A. Battista, A. J. Baird, L. A. Miller
Integrative and comparative biology 55.5 (2015) pp. 901–911 Oxford University Press

[4] **A Full-Body Model of Burn Pathophysiology and Treatment Using the BioGears Engine**
M. McDaniel, A. Baird
2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (2019) pp. 261–264 IEEE

[5] **Open Source Pharmacokinetic/Pharmacodynamic Framework: Tutorial on the BioGears Engine**
M. McDaniel, J. Carter, J. M. Keller, S. A. White, A. Baird
CPT: pharmacometrics & systems pharmacology 8.1 (2019) pp. 12–25 Wiley Online Library

[6] **A Whole-Body Mathematical Model of Sepsis Progression and Treatment Designed in the BioGears Physiology Engine**
M. McDaniel, J. Keller, S. White, A. Baird
Frontiers in physiology 10 (2019) p. 1321 Frontiers

[7] **BioGears: A C++ library for whole body physiology simulations**
A. Baird, M. McDaniel, S. A. White, N. Tatum, L. Marin
Journal of Open Source Software 5.56 (2020) p. 2645

[8] **BurnCare tablet trainer to enhance burn injury care and treatment**
A. Baird, M. Serio-Melvin, M. Hackett, M. Clover, M. McDaniel, M. Rowland, A. Williams, B. Wilson
BMC Emergency Medicine 20.1 (2020) pp. 1–10 BioMed Central

[9] **A multiscale computational model of angiogenesis after traumatic brain injury, investigating the role location plays in volumetric recovery**
A. Baird, L. Oelsner, C. Fisher, M. Witte, M. Huynh
Mathematical Biosciences and Engineering 18.4 (2021) pp. 3227–3257

[10] **Whole Body PBPK Model of Nasal Naloxone Administration to Measure Repeat Dosing**

A. Baird, S. White, R. Das, N. Tatum, E. Bisgaard
Nature Communications - Medicine (2024)

[11] **Increasing Realism and Variety of Virtual Patient Dialogues for Prenatal Counseling Education Through a Novel Application of ChatGPT: Exploratory Observational Study**
G. M, B. A, S. T, J. J. DeBroux, T. Bartlett, M. Krick, U. R
JMIR Med Educ 10 (2024)

Second Section: Collaborative Authorship

None

Third Section: MedEDPortal

None

Fourth Section: Book Chapter

[1] **Numerical Study of Scaling Effects in Peristalsis and Dynamic Suction Pumping**
A. Baird, T. King, L. Miller
Contemp. Math 628 (2014) pp. 129–148

Fifth Section: Published Books, Videos, Software, ect.

[1] **ajbaird/TBISimulator: Alpha 1.0.1 Windows executable and data**
A. Baird
(Sept. 2020) Zenodo

[2] **BioGears: A C++ library for whole body physiology simulations**
A. Baird, M. McDaniel, S. A. White, N. Tatum, L. Marin
(Dec. 2020) Zenodo

[3] **BurnCare Physiology Data: Burn with Resuscitation**
N. Tatum, A. Baird, S. White, M. Clover, M. Hackett
(June 2021) Zenodo

[4] **BurnCare Physiology Data: Burn with Resuscitation and Escharotomy**
N. Tatum, A. Baird, S. White, M. Clover, M. Hackett
(Aug. 2021) Zenodo

[5] **BurnCare Physiology Data: Burn without Intervention**
N. Tatum, A. Baird, S. White, M. Clover, M. Hackett
(Feb. 2021) Zenodo

Sixth Section: Other PublicationsConference Proceedings

[1] **Electro-Dynamic Suction Pumping at Small Scales**
A. Baird, L. Miller
APS Division of Fluid Dynamics Meeting Abstracts, 2013

Invited Presentations

Integrative and Mechanical Physiology Group — *Chapel Hill, NC*
Pumping Mechanisms and their Scaling Effects in Tubular Hearts — *September 2012*

Society of Integrative and Comparative Biology — *San Francisco, CA*
Scaling and Pumping in Tubular Hearts — *January 2013*

Integrative and Mechanical Physiology Group — *Chapel Hill, NC*
Modeling Neurons — *March 2013*

Society of Mathematical Biology — *Osaka, Japan*
Electro-Dynamic Suction Pumping at Small Scales — *August 2014*

Duke Interdisciplinary Discussion Course — *Durham, NC*
Moving Fluid in Tubes — *October, 2014*

Experimental Biology — *Boston, MA*
Implications of Increase Renal Venous Pressure for Renal Hemodynamic and Reabsorptive Function Studied by a Mathematical Model of the Kidney — *March 2015*

Chemical and Biological Defense Science and Technology Conference — *Long Beach, CA*
BioGears: Simulating Whole-Body Response to Chemical Exposure — *November 2017*

International Meeting on Simulation in Healthcare — *Los Angeles, CA*
An In-Silico Whole-Body Framework to Simulate Kinetics and Dynamics of Pharmaceuticals and Associated Reversal Agents — *January 2018*

Virtual Physiological Human Conference — *Zaragoza, Spain*
An In-Silico Whole-Body Framework to Simulate Kinetics and Dynamics of Pharmaceuticals and Associated Reversal Agents — *September 2018*

Department of Defense Working Group on Computational Modeling of Human Lethality, Injury, and Impairment from Blast-Related Threats — *Arlington, VA*
BioGears Human Physiology Engine — *February 2019*

American College of Surgeons Simulation Summit — *Chicago, IL*
BioGears: A Framework for Multiscale Physiology Modeling — *March 2019*

Society for Simulation in Healthcare — *Raleigh, NC*
BioGears Model to Simulate Patient Responses to Sepsis — *March 2019*

BioGears Conference — *Raleigh, NC*
"BioGears Drug Modeling Overview" — *March 2020*
- *Conference Talks:* link

University of Arizona Modeling and Computation Seminar — *Tucson, AZ*
"A Whole-Body Physiological Model of Sepsis and Associated Treatments, Designed in the BioGears Engine" — *March 2021*

University of Arizona Modeling and Computation Seminar *Tucson, AZ*
"A Whole-Body Physiological Model of Sepsis and Associated Treatments, Designed in the BioGears Engine" *March 2021*

CyberPatient Technical Interest Meeting *Virtual*
"BioGears: Physiological Modeling and Engine Construction for High Fidelity Simulation Training" *Jan 2022*

University of Washington Biological Engineering Design Seminar *Seattle, WA*
"Engineering Design with Biological Purpose" *April 2022*

University of Washington Biological Engineering Seminar *Seattle, WA*
"BioGears: Whole-Body Physiological Model to Support Healthcare Simulation, Research and Training" *April 2022*

Society of Industrial and Applied Mathematics: Life Sciences Conference *Pittsburgh, PA*
"Modeling the whole-body response to infection and associated acute inflammation, investigating clinical treatments and outcomes" *July 2022*

Healthcare Systems Modeling and Simulation Quarterly Meeting *Virtual*
"BioGears: Whole-Body Physiological Model to Support Healthcare Simulation, Research and Training" *August 2023*

University of Washington Simulation Symposium *Seattle, Wa*
"Artificial Intelligence and Computer Models for Simulation" *September, 2023*

UW Medicine, Department of Anesthesia Grand Rounds *Seattle, Wa*
"Discussion of Advanced Open-Source Research Tools to Construct and Evaluate the Advanced Joint Airway Management System (AJAMS)" *November 2023*

Other Employment

Applied Research Associates, Inc. *Raleigh, NC*
Biomedical Modeling Group Leader (Senior Engineer, Distinguished Member of the Technical Staff) *February 2016 - October 2021*

- Lead a multidisciplinary team across 4 different projects
- In charge of agile development processes, product roadmap, delivery scheduling, and direct communication with government customers
- Led and won multiple research and development funds through Defense Health Agency and Army Research Lab grants
- Principal investigator of the BioGears, BurnCare training application, and the traumatic brain injury angiogensis projects
- Organized teaming across three research hospitals and multiple small businesses
- Communicate research progress through multiple conferences and peer reviewed publications, including the BioGears 2020 conference
- Oversaw implementation of all models associated with BioGears releases 7.0-7.5